Can Willingness to Pay be Identified without Price Variation?

What Big Data on Usage Tracking Can (and Cannot) Tell Us

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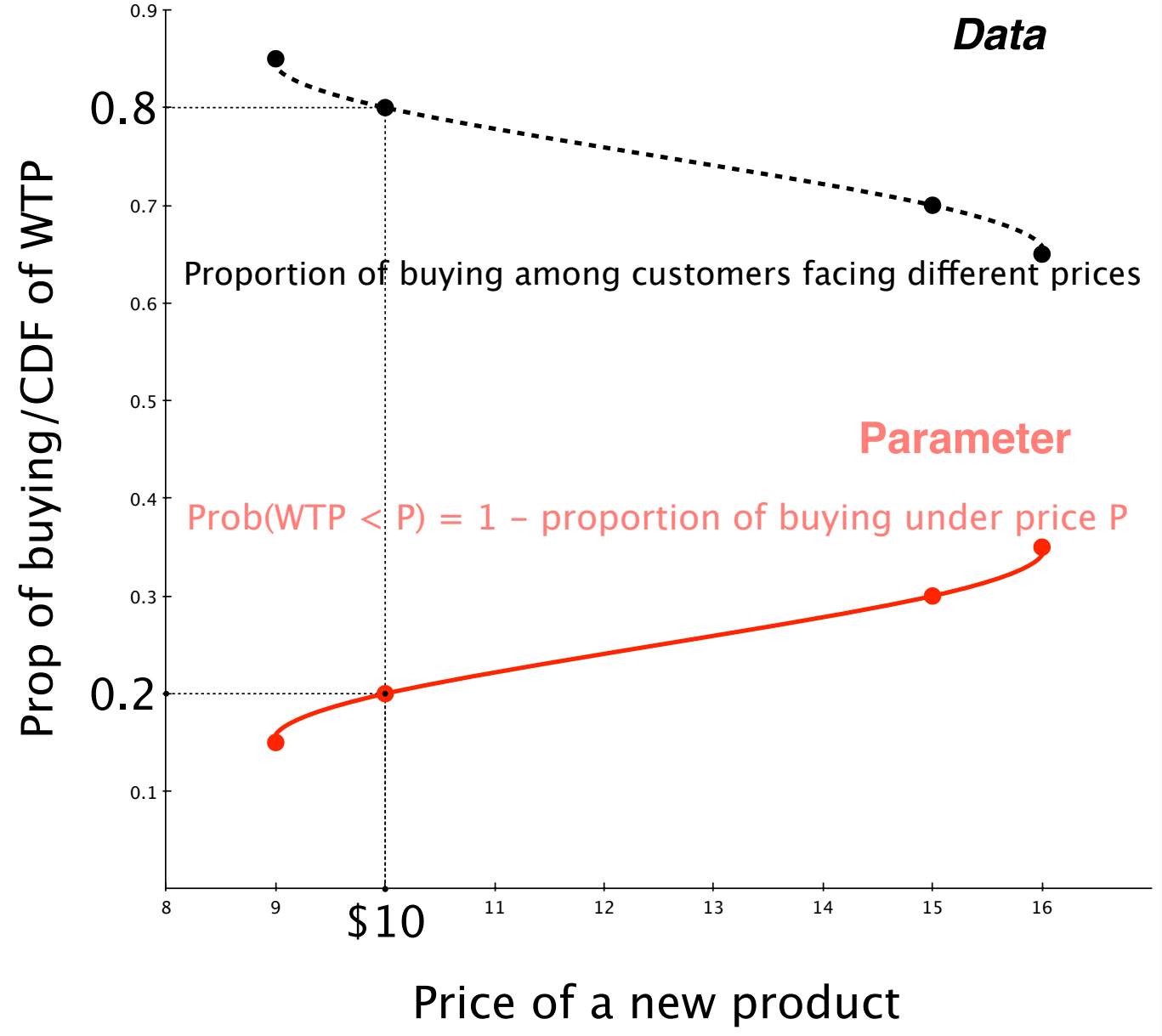
Willingness to Pay (WTP) in marketing

- WTP is essential for customer-driven marketing strategy and marketing mix.
  - market targeting;
  - customer value-based pricing;
  - estimating demand curve;
  - product and service decisions (quality, features, etc.)

Current practice in recovering WTP from data

- Identify the distribution of WTP in a market *with price variation*.
  - Source of price variation: segmented pricing, promotional pricing etc.
  - Current method: **revealed preference** (discrete choice model, auction, Guadagni and Little, 1983, Danthurebandara et al., 2011, Train and Weeks, 2005, Lewbel et al., 2011), and **conjoint analysis** (Green and Rao, 1971, Green and Srinivasan, 1978, Rao, 2014)
- **Price variation is essential for nearly the entire literature.**

Example (Test marketing for a new product). Suppose a consumer buys the product if and only if his or her WTP is higher than price P.



Subscription business

- Subscription market is fast growing and huge
  - Growth rate $> 100\%$ each year in the past 5 years
  - Multibillion revenue per year
- Interested in WTP for subscription service
  - **Challenge: absence of price variation.**
  - Opportunity: big usage tracking data.
  - Why not run price experiment? (Ariely, 2010)
- Research questions:
  - *Can Willingness to Pay be Identified without Price Variation?*
  - Additional value of price variation?

Key insight from a simple example

- How does a firm recover the WTP distribution without Price variation?
- Usage variation $\Rightarrow$ variation in **price of one unit of usage**

User	# Gym Visits Per Month	Monthly Fee	"Price" Per Visit
1	14	\$42	\$3
2	7	\$42	\$6

- (WTP for Gym Membership) = (WTP per [Average] Visit) $\times$ (Expected # Visits).
- Subscribe if (WTP per Visit) $>$ ("Price" Per Visit).
- Market share is informative for distribution of (WTP per Visit).

- Suppose only two usages 7 and 14 visits in population with equal probabilities.

Users visiting 7 times	*Users visiting 14 times*
"Price" per visit $= \$6$	"Price" per visit $= \$3$
Retention rate $= 60\%$	Retention rate $= 80\%$
P(WTP per visit $< \$6) = 0.4$	P(WTP per visit $< \$3) = 0.2$

$$
\begin{aligned}
&P(\text{WTP for Gym Membership} < \$42) = \\
&\qquad P(\#\ \text{Visits} = 7) \times P(\text{WTP per visit} < \$6) + \\
&\qquad P(\#\ \text{Visits} = 14) \times P(\text{WTP per visit} < \$3) = \\
&\qquad\qquad 0.5 \times 0.4 + 0.5 \times 0.2 = 0.3.
\end{aligned}
$$

Steps in finding the distribution of WTP in general

Step 0.	Collect data: (a) subscription/churn choices, (b) usage, (c) product attributes, (d) consumers characteristics.		
Step 1.	Market share or retention rate by usage	$\Rightarrow$	Distribution of WTP per unit of usage
Step 2.	Usage data	$\Rightarrow$	Distribution of expected usage
Step 3.	Combine the distribution of WTP per visit and usage	$\Rightarrow$	Distribution of WTP for service
Step 4.	Estimate elasticities, demand curve, and do counterfactual		

Why is it challenging?

- Heterogeneity
  - WTP per visit
  - usage (expected # of visits)
- Correlation between WTP per visit and usage
- Unobserved usage by nonsubscribers
- Selection: nonsubscribers are different from subscribers in unobservable ways

Our approach accomodates all the above issues.

Contribution

- Main contribution: a novel method to identify & estimate semiparametrically the conditional distribution of WTP given customer characteristics and product features **when only usage variation is present**.
- No research that demonstrates how to obtain the WTP distribution in the **absence of price variation**.
- Closest reserach: Nevo et al. (2016).

Big usage data of YBOX, a music streaming service

- YBOX is a music streaming service targeting Southeast Asia. 80% market share, $>$ 10 million users.
- 1 million users data (Jan 2015–Feb 2017):
  - subscription history
  - daily # of seconds listening music with the service
  - basic demographics (age and gender)
- **Three prices** (*roughly* \$5 [56.8%], \$4 [17.6%], \$3 [23.9%]) for the same monthly music streaming service
- Average monthly listening hours range from less than 1 hour to more than 150 hours.

Descriptive statistics of YBOX

Table: Descriptive Stat YBOX Users

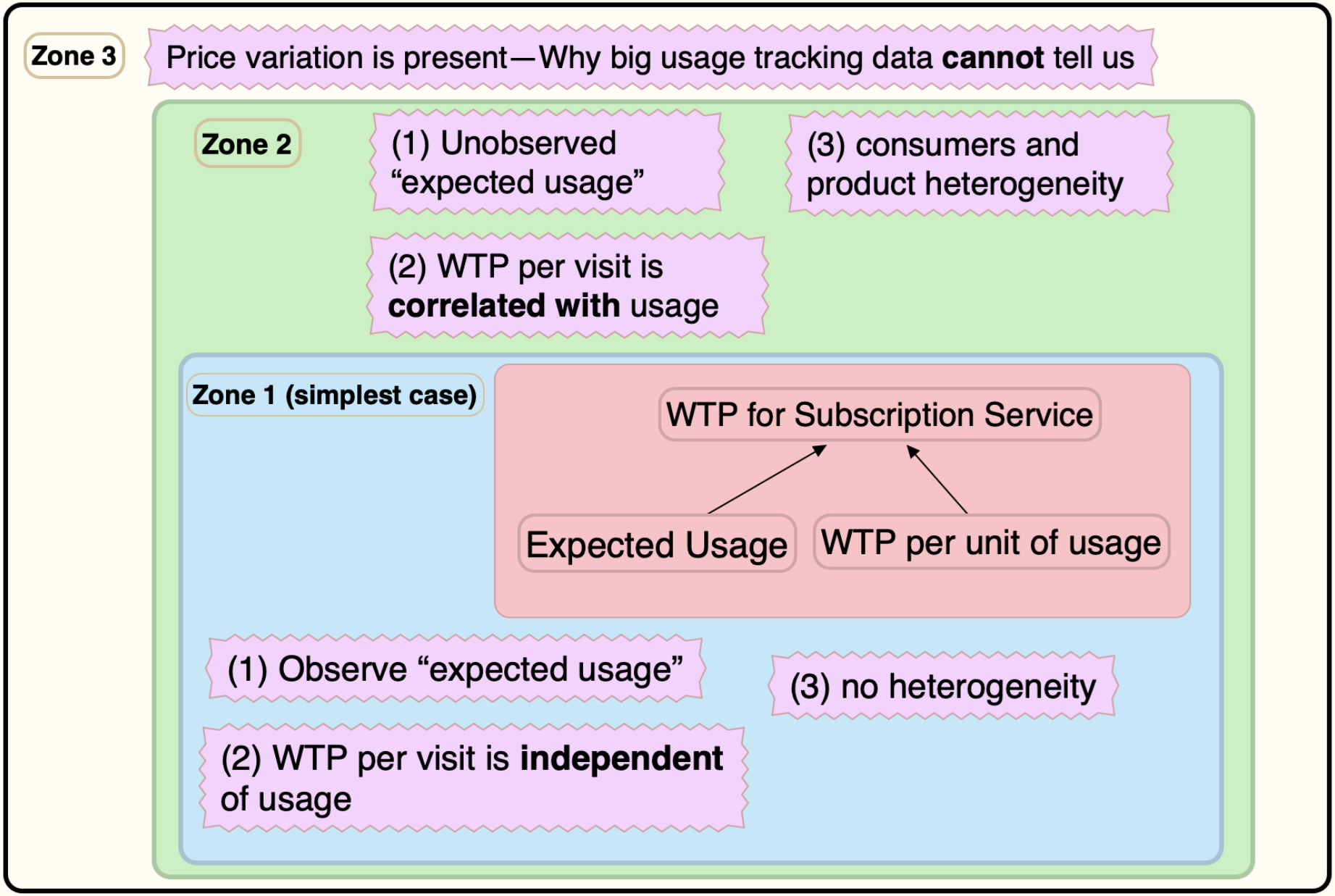
	Users churned	Users not churn
Average daily listening hours	1.17	1.33
	(0.99)	(1.09)
Age	29.53	29.92
	(8.25)	(8.14)
Is male	0.51	0.55
n of obs	1211	7487

- Randomly sampled 8698 users from one city.

Estimated WTP for YBOX monthly plan

	Estimated Mean of Log WTP	
Age group	Men	Women
Before college	log($16)	log($16)
In college	log($24)	log($21)
23–30	log($33)	log($30)
$>$ 30	log($49)	log($56)

Outline


Zone 3
Price variation is present—Why big usage tracking data **cannot** tell us
Zone 2
(1) Unobserved "expected usage"
(3) consumers and product heterogeneity
(2) WTP per visit is **correlated with** usage
Zone 1 (simplest case)
WTP for Subscription Service
Expected Usage
WTP per unit of usage
(1) Observe "expected usage"
(3) no heterogeneity
(2) WTP per visit is **independent** of usage

Model setup: simplest case (ZONE 1)

Notation:

- i indicates a consumer
- WTP for **service**: W_i
- WTP **per unit of usage**: α_i
- Expected usage: Q_i^*
- Decision: $S_i = 1$ (sub) and $= 0$ (not)

Assumption

1. $W_i = \alpha_i Q_i^* \Rightarrow \ln W_i = \ln \alpha_i + \ln Q_i^*$.
2. Q_i^* is observed for both subscribers and nonsubscribers.
3. $\alpha_i \perp\!\!\!\perp Q_i^*$.

These will be relaxed.

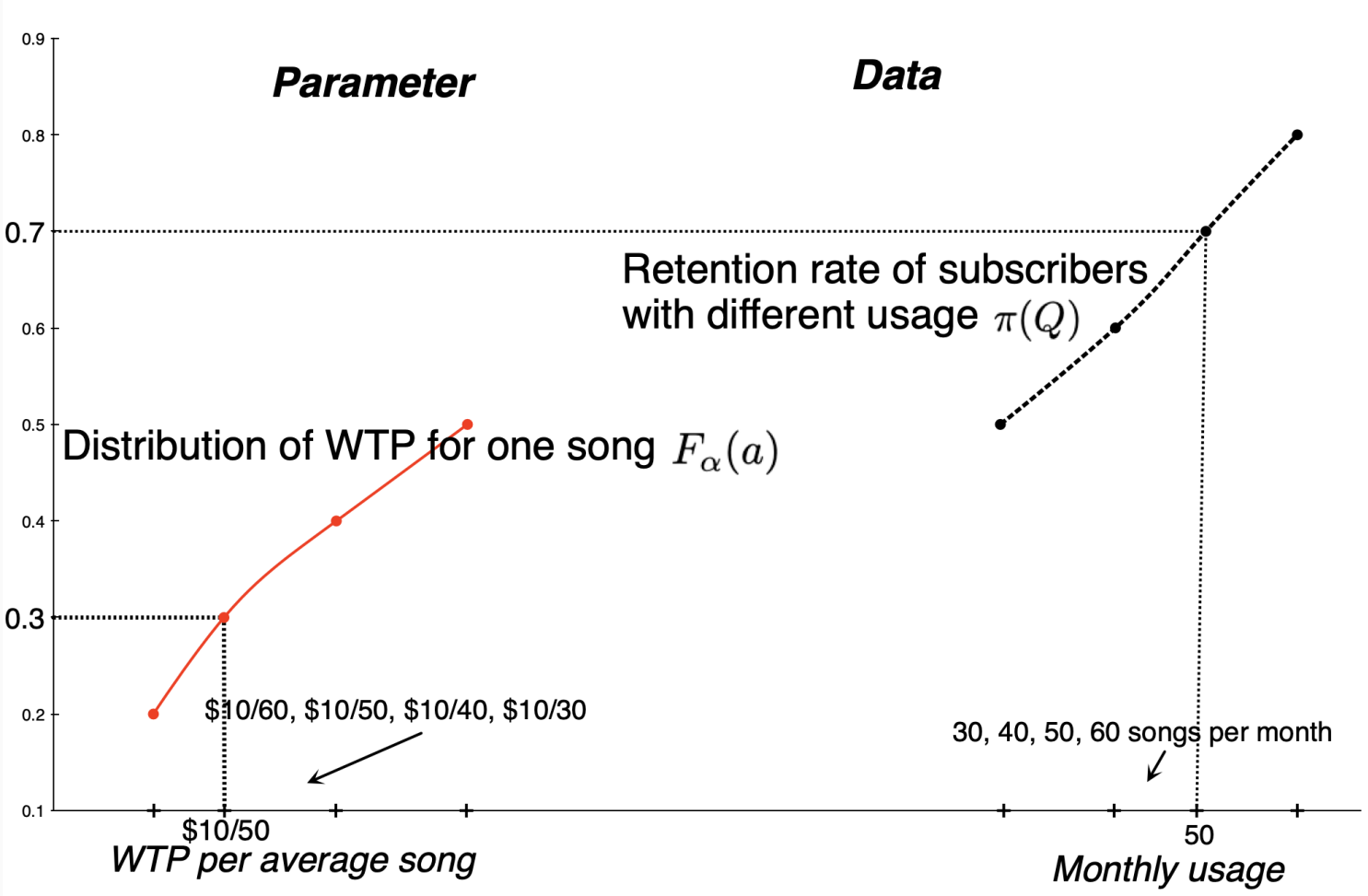
▸ Microfoundation of usage

Decision rule: $S_i = \mathbb{I}(P < W_i) = \mathbb{I}(P < \alpha_i Q_i^*)$.

Nonparametric estimation of $F_\alpha(\alpha_i)$ (**Step 1**)

Decision rule $S_i = \mathbb{I}(P < \alpha_i Q_i^*)$ implies that

$$F_\alpha(P/Q) \equiv \Pr(\alpha_i \leq P/Q) = 1 - \underbrace{\Pr(S_i = 1 \mid Q_i^* = Q)}_{\pi(Q)} \quad \Rightarrow \quad \boxed{F_\alpha(a) = 1 - \pi(P/a)}$$



WTP: From usage $\Rightarrow$ subscription (**Step 2 and 3**)

- If
  - $F_\alpha(\alpha_i)$ is known,
  - Q_i^* is observed, so $F_Q(Q_i^*)$ is known, (**Step 2**)
  - $\alpha_i \perp\!\!\!\perp Q_i^*$,

  the distribution $F_W(W_i)$ is calculated by using

$$W_i = \alpha_i Q_i^*.$$

- It can shown that

$$\boxed{F_W(w) = 1 - \mathrm{E}\Big(\pi\Big(\frac{PQ_i^*}{w}\Big)\Big).}$$

See derivation

Estimation algorithm

Algorithm for Zone 1

Step 1: estimate $\pi(q) \equiv \Pr(S_i = 1 \mid Q_i^* = q)$;

Step 2a: estimate $\hat{F}_\alpha(a) = 1 - \hat{\pi}(P/a)$.

Step 2b: estimate $\hat{F}_W(w) = 1 - n^{-1} \sum_{i=1}^n \hat{\pi}(PQ_i^*/w)$, because

$$\begin{aligned} F_W(w) &= 1 - \mathrm{E}(\pi(PQ_i^*/w)) \\ &\approx 1 - n^{-1} \sum_{i=1}^n \hat{\pi}(PQ_i^*/w). \end{aligned}$$

Applications and counterfactual analysis (**Step 4**)

- Demand curve: $D(p) \equiv \Pr(S_i = 1 \mid P = p) = 1 - F_W(p)$.
- Pricing a shorter subscription plan based on **higher frequency usage** data.
  - Let $Q^*_{i,wk}$ be the expected usage in *week* $wk = 1, \ldots, 4$.

$$W_{i,wk} = \alpha_i Q^*_{i,wk}.$$

  - Note that in general $Q^*_{i,wk}$ and $Q^*_i/4$ (Q^*_i is monthly usage) have different distributions.
- More possible counterfactual analysis in Zone 2.

Summary by R practice

- https://chengchou.shinyapps.io/WTPV1/

Estimated elasticities of WTP for YBOX monthly plan

- Age elasticity: 0.04 (0.01).
- Men are willing to pay 26.23% (18.09%) more than women.
- For Age > 30, the mean monthly listening hours of men and women are 19.6 and 21.5 hours.

	Estimated Mean of Log WTP	
Age	Men	Women
Before college	log($16)	log($16)
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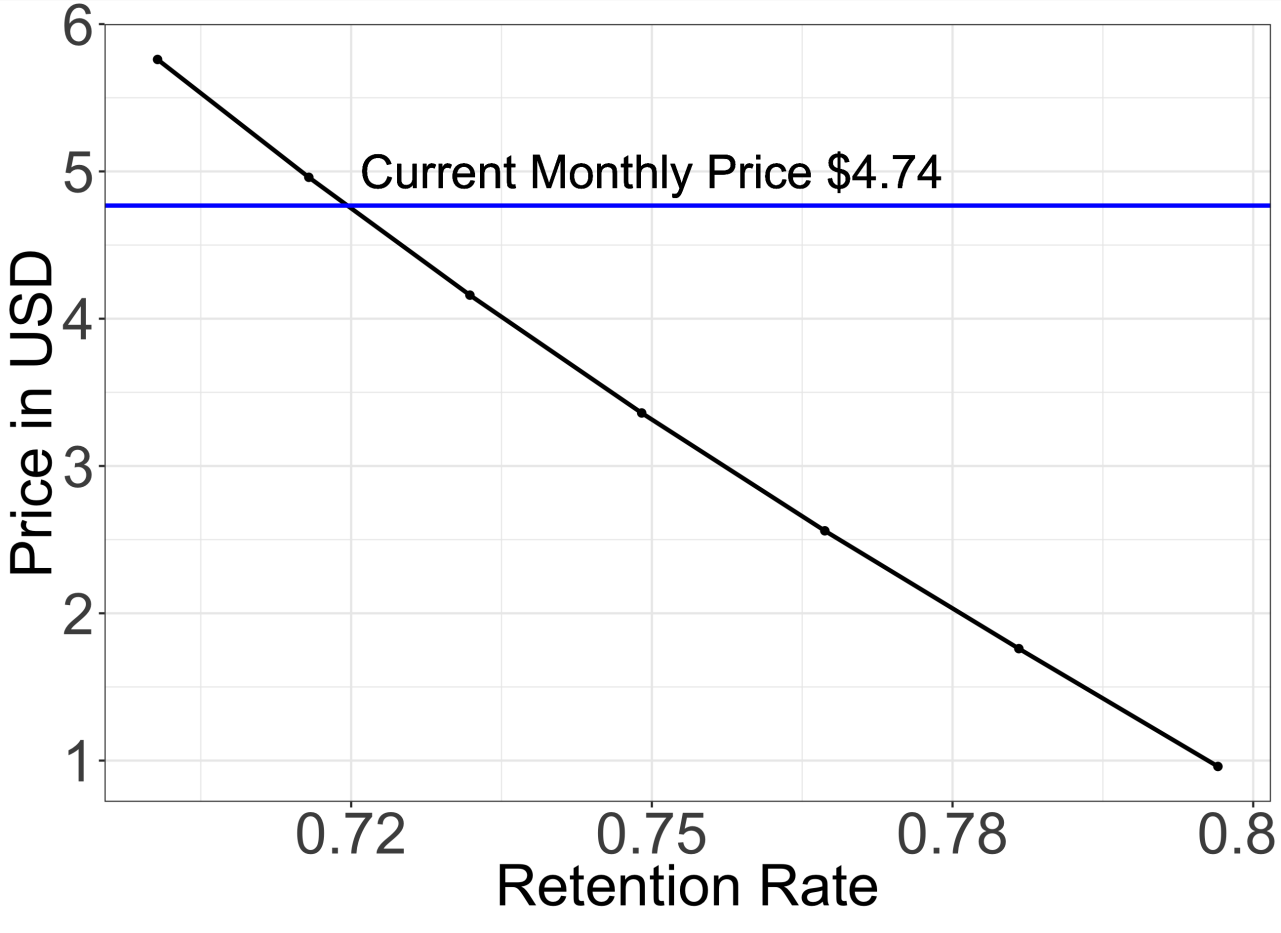
> 30	log($49)	log($56)

Estimated demand curve for current **monthly** streaming plan

Inelastic. Is increasing the current price profitable? Depends on marginal cost, but probably YES.

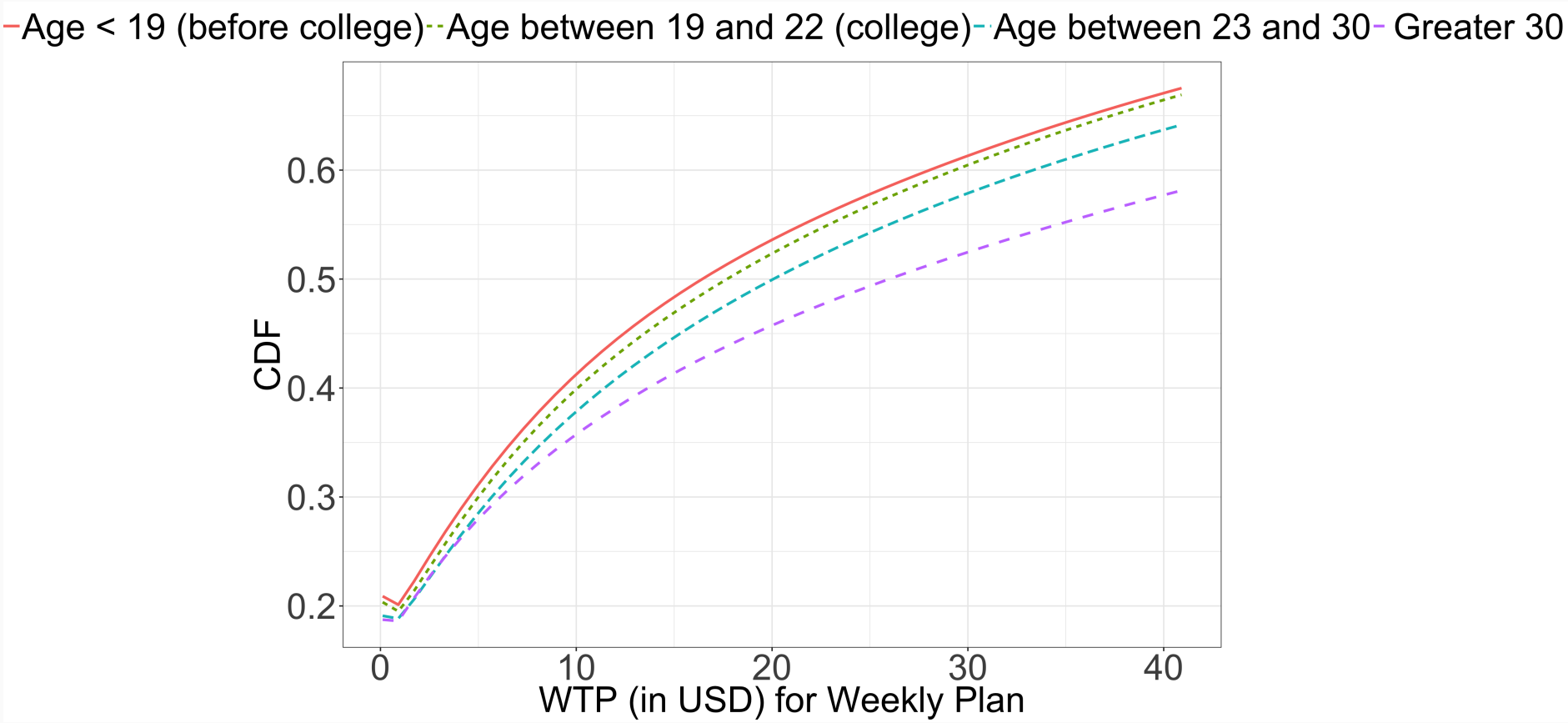
Demand for **counterfactual weekly** streaming plan



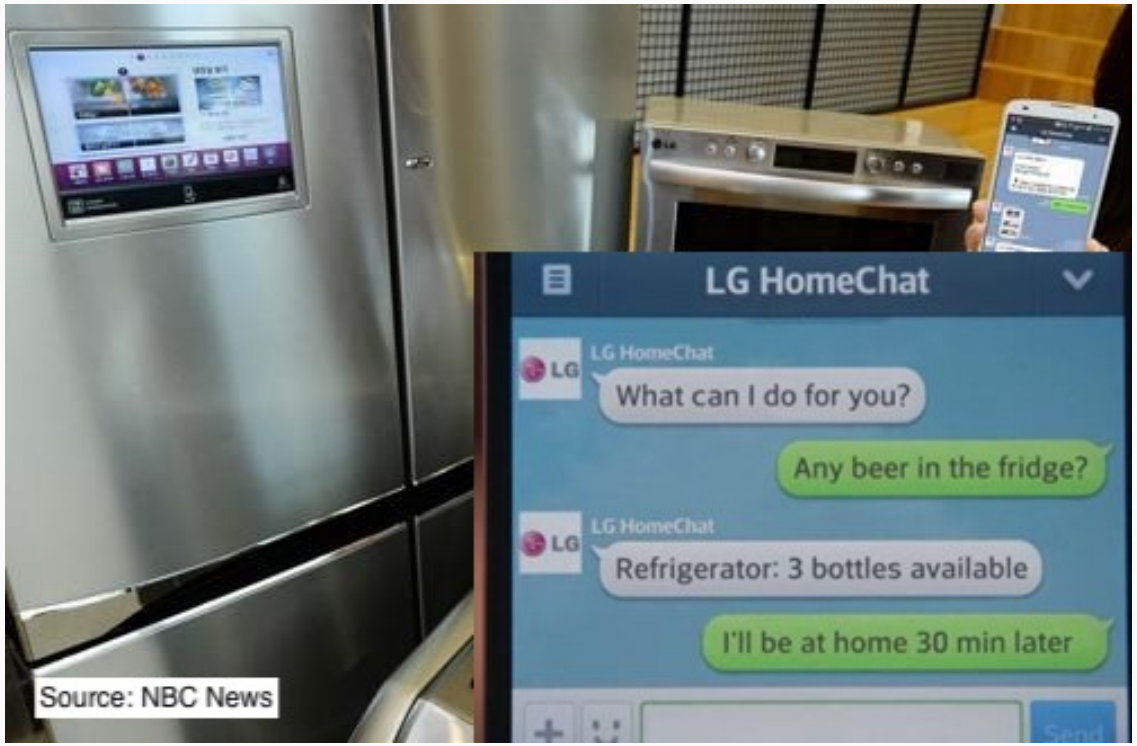
To obtain 75% retention rate,

- the monthly price is \$8.8,
- the weekly price is \$3.2 HIGHER than \$2.2 = \$8.8 / 4.

Estimated distribution of the WTP for counterfactual weekly plan



Conclusion: a bigger picture (of a fridge)



- Essentially, we need the separation of purchase (subscription) and consumption (usage).
- Such separation also holds in packaged goods (beer)—but we did not track the usage.
- 5G and Internet of Things could enable the tracking.

ZONE 2 and 3.

Benefits of ZONE 2: conditional distribution of WTP $F_W(W_i \mid X_i)$

- X_i: observed consumer and product characteristics
- $F_W(W_i \mid X_i) \Rightarrow$ elasticities
- **Elasticities** (we derived explicit formulas): (**Step 4**)

$$\mathrm{E}\left(\frac{\partial \mathrm{E}(\ln W_i \mid X_i)}{\partial X_{j,i}}\right), \text{ e.g., } X_{j,i} \text{ is ad. spending}$$

$$\mathrm{E}\Big[\mathrm{E}(\ln W_i \mid X_{other}, \text{Play music offline} = \text{Yes}) - \mathrm{E}(\ln W_i \mid X_{other}, \text{Play music offline} = No)\Big]$$

Two dimensional heterogeneity (**Step 1 and 2**)

Heterogeneity in WTP per song.

- Project unobserved $\ln \alpha_i$ on X_{1i}:

$$\ln \alpha_i = \beta' X_{1i} + U_i.$$

- Estimate β and $F_U(U_i)$ **without parametric assump.** for U_i.
- Implies $S_i = \mathbb{I}(\beta' X_{1i} + \ln Q_i^* - \ln P + U_i > 0)$.

Theorem

Under certain support conditions, β can be consistently estimated by running OLS of

$$\frac{S_i - \mathbb{I}(\ln Q_i^* - \ln P > 0)}{\underbrace{f_Q(Q_i^* \mid X_{1i})}_{\text{Nonparametric Estimator}}}.$$

on X_{1i}.

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Required Assumptions

Assumptions

1. Q_i^* is observable;
2. **Valid and relevant IV:** $\mathrm{E}(Z_i U_i) = 0$, $\mathrm{E}(Z_i Z_i')$ is nonsingular, and $\operatorname{rank} \mathrm{E}(X_{1i} Z_i') = \dim(X_{1i})$;
3. **Independence of consumption:** $Q_i^* \perp\!\!\!\perp U_i \mid (X_{1i}, Z_i)$;
4. **Big support of consumption:** the support of $\ln Q_i^* \mid (X_{1i}, Z_i)$ covers the support of $\underbrace{(\ln P - \beta' X_{1i} - U_i)}_{=\ln(P/\alpha_i)} \mid (X_{1i}, Z_i)$.

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Heterogeneity in usage (# of songs), unobserved usage by nonsubscribers, and selection

- Reduced form of Q^*_{it}:

$$\ln Q^*_{it} = \gamma' X_{2it} + V_i \qquad \text{for subscribers and nonsubscribers}$$

$$\ln Q_{it} = \ln Q^*_{it} + \varepsilon_{it} \qquad \text{for subscribers}$$

  V_i and U_{it} (hence α_{it}) are correlated (selection)

- **Panel data:** Observed log usage Q_{it} for subscribers, and X_{2it} for both subscribers and nonsubscribers.
- Fixed effect reg $\Rightarrow \gamma$ and **"predicted usage"** $\bar{Q}_{it} = \exp(\gamma' X_{2it})$ for both subscribers and nonsubscribers. Time invariant characteristics
- Identify the distribution of $\eta_{it} \equiv (U_{it} + V_i) = \ln W_{it} - (\beta' X_{1it} + \gamma' X_{2it})$.

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- Identify the distribution of $\eta_{it} \equiv (U_{it} + V_i) = \ln W_{it} - (\beta' X_{1it} + \gamma' X_{2it})$.

- Using $\ln \bar{Q}_{it} = \gamma' X_{2it}$, we have

$$S_i = \mathbb{I}(\beta' X_{1it} + \ln \bar{Q}_{it} - \ln P + (U_{it} + V_i) > 0).$$

- Estimate β similarly.
- **Excluded variable requirement:** need variables that affect usage but not the WTP per unit of usage, e.g. weather.
- Why? **Support condition:** the support of $\ln \bar{Q}_{it} \mid (X_{1it}, Z_{it})$ covers the support of $(\ln P - \beta' X_{1it} - U_{it} - V_i) \mid (X_{1it}, Z_{it})$.

Obtaining the conditional WTP

Theorem: Conditional Distribution of WTP

- Obtain β as 2SLS estimator of regressing Y_{2it} on X_{1it} with IV Z_{it}

$$Y_{2it} = \frac{S_{it} - \mathbb{I}(\ln \bar{Q}_{it} - \ln P > 0)}{f_{\bar{Q}}(\bar{Q}_{it} \mid X_{1it}, Z_{it})}.$$

- Obtain the conditional WTP distribution:

$$\mathrm{E}(\ln W_{it} \mid X_{1it}, X_{2it}) = \beta' X_{1it} + \gamma' X_{2it} + \mathrm{E}(\eta_{it} \mid X_{1it}, X_{2it})$$

$$\eta_{it} = \mathrm{E}\left(\frac{S - \mathbb{I}(\ln P - \beta' X_{1it} - \ln \bar{Q}_{it} \leq 0)}{f_{\bar{Q}_{it}}(\bar{Q}_{it} \mid X_{1it}, Z_{it})} \middle| X_{1it}, Z_{it} \right)$$

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Counterfactual (**Step 4**)

- Evaluate the money-metric effect of product changes
- Pricing based on usage

BACKUP SLIDES

Include time invariant X_{2it}

- Define $\eta_{it} = U_{it} + V_i$. Given $\ln \bar{Q}_{it} = \gamma' X_{2it}$, we have

$$S_{it} = \mathbb{I}(\beta' X_{1it} + \ln \bar{Q}_{it} - \ln P + \eta_{it} > 0).$$

  Then β can be estimated using the stated theorem.

- Suppose $X'_{2it} = (X'_{2i,a}, X'_{2it,b})$. We still write

$$\ln Q^*_{it} = \gamma'_a X_{2i,a} + \gamma'_b X_{2it,b} + V_i.$$

  We cannot identify γ_a, but can identify γ_b.

- Define $\ln \bar{Q}_{it} = \gamma'_b X_{2it,b}$, and we have

$$S_{it} = \mathbb{I}(\beta' X_{1it} + \gamma'_a X_{2i,a} + \ln \bar{Q}_{it} - \ln P + \eta_{it} > 0).$$

  When X_{1it} and $X_{2i,a}$ are different, we can separately identify β and γ_a.

Back to 'usage heterogeneity'

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◂ Back to 'usage heterogeneity'

$$
\begin{aligned}
F_W(w) &= \Pr(W_i \leq w) \\
&= \Pr(\alpha_i Q_i^* \leq w) \\
&= \int_0^\infty \Pr(\alpha_i q_i \leq w) \, \mathrm{d}\, F_Q(q_i) \\
&= \int_0^\infty F_\alpha(w/q_i) \, \mathrm{d}\, F_Q(q_i) \\
&= \mathrm{E}(F_\alpha(w/Q_i^*)) \\
&= 1 - \mathrm{E}(\pi(PQ_i^*/w)).
\end{aligned}
$$

Back

Usage Model: Microfoundations

- Earlier with exogenous Q_i^* - the usage model was not specified.
- Each subscription setting might have a different micro-model
- Spotify Example: Consumers solve time allocation problem between listening to music and the outside option.
  - $u(q, \nu, T; \theta) = \theta_1 q \underbrace{\nu}_{\text{shock}} + \theta_2 q^2 + (\underbrace{T}_{\text{Time budget}} - q)$
  - Obtain the usage decision rule: $q(\nu, T; \theta)$
  - $Q_i^* = E_\nu [q(\nu, T; \theta)]$
- Issue to consider: Does θ (parameters of micromodel) depend on the counterfactual?
  - If it does not: we can use the same micromodel in counterfactuals. Don't really need to estimate θ.
  - if it does: we need to estimate the micromodel parameters θ

Back to simplest model